DTC U3000:49 [i-ELOOP]

id130201001500

Note
- If the capacitor (i-ELOOP) voltage is low after replacing the DC-DC converter (i-ELOOP), the i-ELOOP indicator light (green) flashes until the capacitor (i-ELOOP) voltage is **approx. 9.5 V**.

DTC U3000:49	**DC-DC converter (i-ELOOP) internal malfunction**
DETECTION CONDITION	• A malfunction in the DC-DC converter (i-ELOOP) is detected.
FAIL-SAFE FUNCTION	• Inhibits engine-stop by operating the i-stop function. • Switches to bypass mode which shuts the capacitor (i-ELOOP) off.
POSSIBLE CAUSE	• DC-DC converter (i-ELOOP) internal malfunction
SYSTEM WIRING DIAGRAM	• Not applicable

Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
2	**PERFORM DTC INSPECTION AND VERIFY DC-DC CONVERTER (i-ELOOP) MALFUNCTION** • Switch the ignition off, then ON (engine off). • Perform the DC-DC converter (i-ELOOP) DTC inspection using the M-MDS. (See DTC INSPECTION [i-ELOOP].) • Are any DTCs other than DTC U3000:49 displayed?	Yes	DTC P1794:00 is displayed: • Go to the next step. Any other DTCs are displayed: • Go to the applicable DTC inspection. (See DTC TABLE [i-ELOOP].)
		No	Replace the DC-DC converter (i-ELOOP), then go to Step 6. (See DC-DC CONVERTER (i-ELOOP) REMOVAL/ INSTALLATION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See DC-DC CONVERTER (i-ELOOP) REMOVAL/ INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].)
3	**VERIFY THE OPEN CIRCUIT CONDITION OF THE CAPACITOR CUIRCUIT** • Access the CAP_SYS10 PID using the M-MDS. (See PID/DATA MONITOR INSPECTION [i-ELOOP].) • Is the CAP_SYS10 PID value off?	Yes	Reconnect the service plug, then go to the next Step.
		No	Go to the next step.
4	**DETERMINE IF THE MALFUNCTION IS CAUSED BY CAPACITOR (i-ELOOP) BY VERIFYING THE PID RELATED TO THE CAPACITOR (i-ELOOP)** • Access the following DC-DC converter (i-ELOOP) PIDs using the M-MDS (See PID/DATA MONITOR INSPECTION [i-ELOOP].) — CAP_SYS_10 — CAP_SYS_14 — CAP_SYS_16 — CAP_SYS_17 • Are all of the PID item “CAP_SYS10, CAP_SYS_14, CAP_SYS_16, CAP_SYS_17” values off?	Yes	Replace the DC-DC converter (i-ELOOP), then go to the next step. (See DC-DC CONVERTER (i-ELOOP) REMOVAL/ INSTALLATION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See DC-DC CONVERTER (i-ELOOP) REMOVAL/ INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].)
		No	Replace the capacitor (i-ELOOP), then go to the next step. (See CAPACITOR (i-ELOOP) REMOVAL/ INSTALLATION [i-ELOOP].)

STEP	INSPECTION		ACTION
5	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Start the engine and leave for **100 s or more**. • Clear the DTC for the DC-DC converter (i-ELOOP) using the M-MDS. (See CLEARING DTC [i-ELOOP].) • Start the engine. • Leave for **1 min** while racing with high electrical load. (DC-DC converter (i-ELOOP) terminal 2A voltage is **18 V or more**.) • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Remove the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) • Wait for **2 min**. • Reinstall the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) • Reconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Wait for **1 min**. • Switch the ignition ON. • Retrieve the DC-DC converter (i-ELOOP) DTCs using the M-MDS. (See DTC INSPECTION [i-ELOOP].) • Is the same DTC present?	Yes	Replace the DC-DC converter (i-ELOOP). (See DC-DC CONVERTER (i-ELOOP) REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See DC-DC CONVERTER (i-ELOOP) REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].) • Start the engine and leave for **100 s or more**. • Clear the DTC for the DC-DC converter (i-ELOOP) using the M-MDS. (See CLEARING DTC [i-ELOOP].) Go to the next step.
		No	Go to the next step.
6	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [i-ELOOP].)
		No	DTC troubleshooting completed.